Curriculum Development

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Introduction

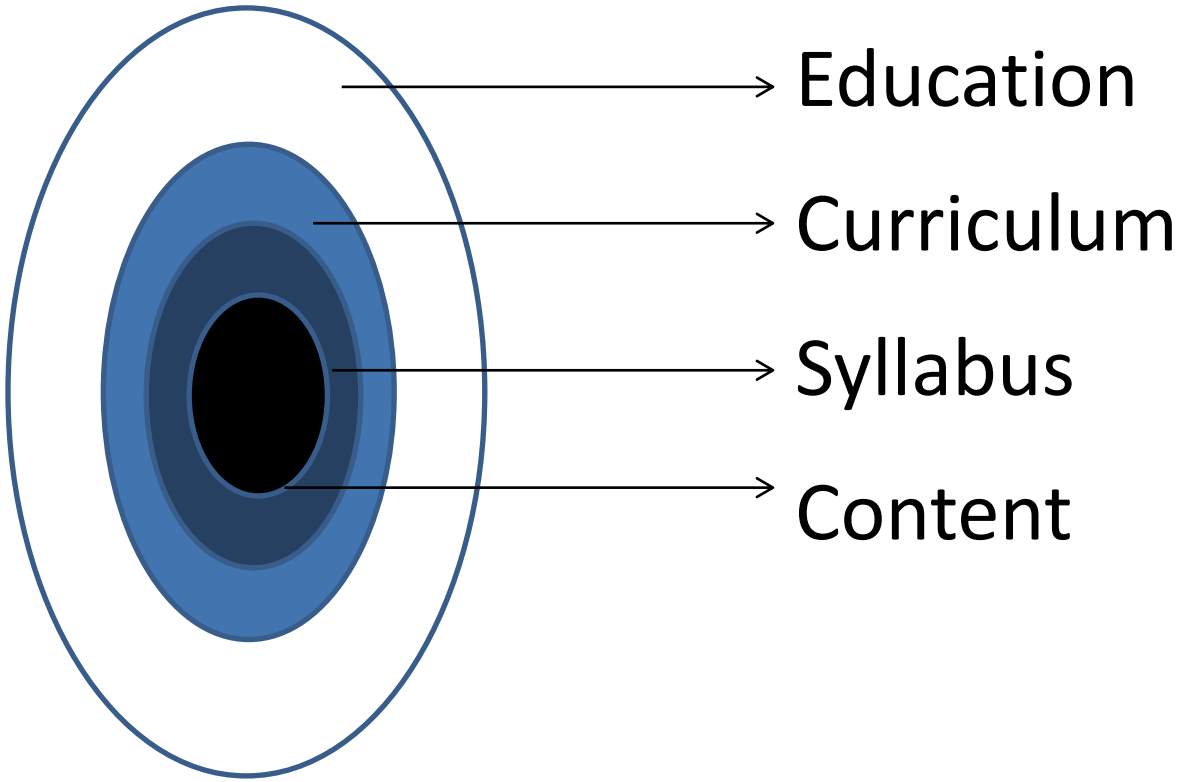
- The word curriculum is derived from the Latin ***currere*** which means ‘to run’.

- Curriculum provides a template or **design** which **enables learning** to take place.

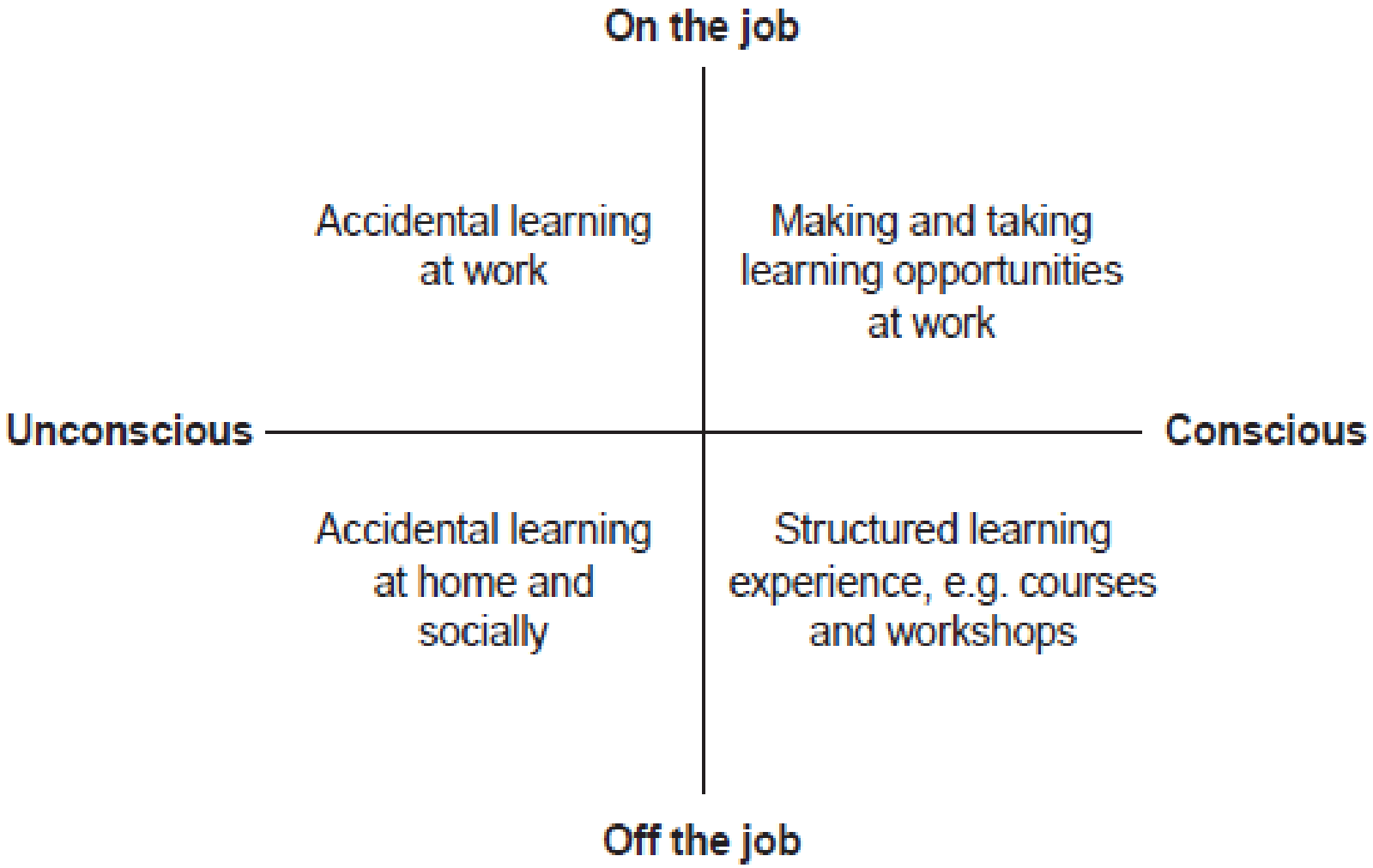
Introduction…..

- A curriculum is more than a syllabus.
- A **syllabus** describes the content of a programme and can be seen as one part of a curriculum.

Introduction……



Sources of learning



Types of Curriculum

- The **formal curriculum**: what the institution plans and organizes to teach

- The **informal curriculum**: what the students learn from a variety of sources and interactions while taking part in the ‘extra-curricular’ activities at the institution

- The **hidden curriculum**: what the students learn but the institution did not set out to teach. Much of this refers to the aspects like ‘caught not taught’.

Some definitions

- **Albert Oliver (1977):** curriculum is “the educational program of the school” and divided into four basic elements: 1) program of studies, 2) program of experiences, 3) program of service, 4) hidden curriculum.

- **Bell (1971):** the offering of socially valued knowledge, skills, and attitudes made available to students through a variety of arrangements during the time they are at school, college, or university.

- **Harnack (1968):** The curriculum embodies all the teaching-learning experiences guided and directed by the school.

Definitions….

- **Ralph Tyler (1957):** The curriculum is all of the learning of students which is planned by and directed by the school to attain its educational goals.

- **David G. Armstrong (1989):** "is a master plan for selecting content and organizing learning experiences for the purpose of changing and developing learners' behaviors and insights."

Bringing all these points together, the ***curriculum*** *is viewed as a composite whole including the learner, the teacher, teaching and learning methodologies, anticipated and unanticipated experiences, outputs and outcomes possible within a learning institution.*

Ways of viewing curriculum

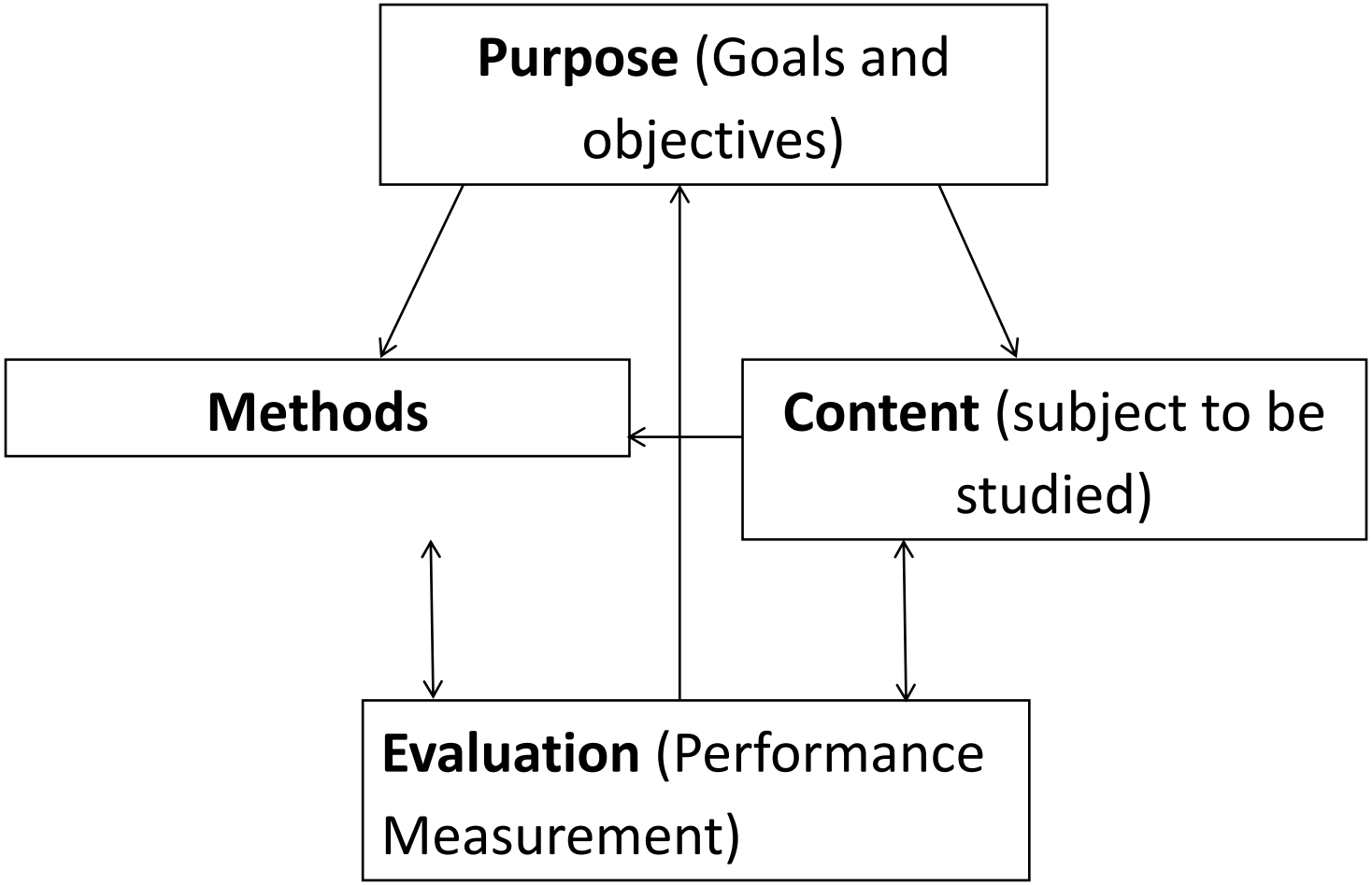
- Curriculum as **a product** - program, document, electronic media, or multimedia
- Curriculum as **a program of study** - usually courses offered, curriculum sequences of study in standards as benchmarks, gateways,
- Curriculum as **intended learnings** - goals, content, concepts, generalizations, outcomes
- Curriculum as **experiences of the learner** - activities, planned and unplanned.
- **Hidden curriculum** - what students learn that isn't planned (also called **unofficial** or **counter**)

Elements of curriculum

The curriculum has four elements that are in constant interaction:

- purpose (goals and objectives)
- content or subject matter
- methods
- evaluation

Interaction among elements



Advantages and limitations of curriculum

Advantages

- Makes learning specific
- Makes learning systemic
- Makes learning measurable and concrete
- Saves time and resources to conduct educational activity
- Helps achieve goals and objectives if planned correctly
- National and international recognition

Limitations

- Society always changing, so difficult to make changes in curriculum accordingly
- Needs intensive work to develop a good curriculum
- Makes education narrow and less open-ended usually
- Can’t include all learning theories
- Curriculum made at one setting may not be applicable in other settings
- Curriculum change requires intense, tedious processes.

Principles of curriculum design

- **Principle of Utility or applicability**
- **Principle of flexibility**
- **Principle of child-centeredness (student)**
- **Principle of life-centeredness**
- **Principle of community-centeredness**
- **Principle of correlation (interdisciplinary)**
- **Principle of activity-centeredness**

Principles…

- **Principle for the use of leisure**
- **Principle of development of culture and civilization**
- **Principle of need based activity**
- **Principle of value (justice, equity, opportunity)**
- **Conservation principles**
  - Conservations of natural resources, wildlife should be inculcated.
- **Principle of creativity**
- **Principle of harmony and balance**

Factors affecting curriculum design

- **Technology** has had a tremendous impact on society.
- Educational curriculums must **keep up with the pace** so that future generations will **benefit**.
- The concept of knowledge has changed from **having information in the brain**, to "having access to information about a particular topic and **knowing how to use it**".
- Hence, the curriculum is determined by a number of factors that can be broadly classified as the **extrinsic and intrinsic factors**.

Determinants

- **Extrinsic**
  - Political
  - Economic
  - Technological
  - Socio-cultural, Religious
  - Philosophical
- **Intrinsic**
  - Acceptance of the curriculum within the institution
  - organizational policy
  - Managerial aspects of the institution
  - Organizational environment etc.

Extrinsic Factors

- **Political**
  - From defining goals, interpreting curricular materials to approving examination systems, politics affects curriculum design.
- **Economic**
  - Curriculum developed for in-house training in corporations focuses on educating employees for promotions that bring better returns in profits.
  - Nations financing on education expect an economic return from educated students contributing to the country's economy with global competition abilities in technical fields.

Extrinsic…

- **Technological**
  - E.g. Means of communication affect curriculum design
- **Diversity (Socio-cultural and religious)**
  - Affected by forces like conflicts, concepts of justice and equity, inclusiveness, ethics etc.
- **Learning theories (Philosophical)**
  - E.g. change in child and adult learning theories affect curriculum design. The contents are selected accordingly.

Intrinsic factors

- **Acceptance** of the curriculum by teachers, students and administrators of the academic institutions also determines curriculum design

- Sound organizational **policy**, flexibility, financial **stability**, administrative **capacity**

Intrinsic…

- **Managerial aspects** of the institution (leadership, communication, coordination, motivation, staffing, budgeting etc.)

- Organizational **environment** (physical infrastructures, safety, peace, harmony, innovative environment for students, flexible laws and orders

Models/Approaches of curriculum design

- Content oriented
- Behavioristic
- Life Process oriented
- Competency based
- Flexner model
- The 'SPICES' model of curriculum design
- PRISMS model

Content oriented

- Ancient model of curriculum design
- Mastery of subject matter is central task
- Standards are set for
  - Subject matter covered
  - Learner’s mastery on the content
- A textbook is the primary instructional tool

- The library is used primarily in content-oriented ways with a focus on selection and use of specific materials in a single content area.
- Interrelations between various subjects are not stressed
- Questions focus on "what" rather than "how" or "why."

Behavioristic

- The nature of a person's thinking can be affected through a variety of **rewards and consequences**.
- The aim for education through behaviorism is to develop **appropriate behaviors** in the classroom setting that are conducive to learning.
- Behaviorism refers more to **conditioning the learner** to be ready to learn and not so much educating.

Behavioristic…

- **Step by step conditioning** programs are used to achieve the desired behavior and **rewards** are used to **motivate**.
- Learners receive simple forms of motivation such as **verbal praise**, **smiles**, a **touch** on the shoulder, and **candy**.
- **Extrinsic** rewards are **followed by** the **intrinsic** rewards

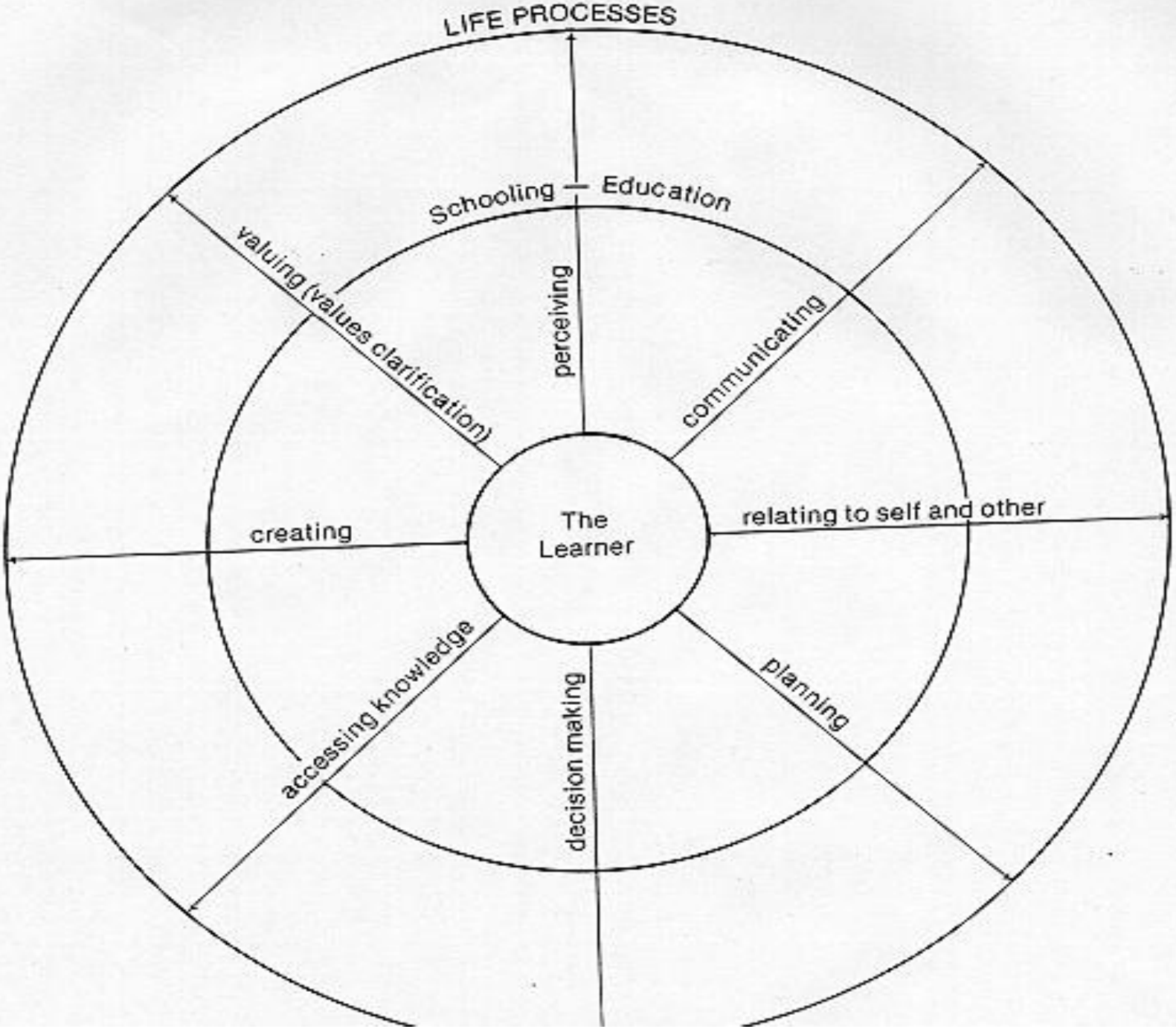
Behavioristic…

- The role of the **teacher** as a behaviorist is a **conditioner**.
- If students can change through their own thinking, they will begin to become **confident** and **believe in themselves** more.

Life Process oriented

- The development of skills and traits that will serve the learner for a **lifetime processes**

- There is a **higher degree** of carry-over into everyday living experiences **than in other designs**


LIFE PROCESSES
Schooling — Education
valuing (values clarification)
perceiving
communicating
creating
The Learner
relating to self and other
accessing knowledge
decision making
planning

Life Process oriented…

- Better **balance** between affective and cognitive considerations.
- The library can provide materials, resources and services promoting **independent learning** and **creativity.**
- This curriculum design suffers from a **difficulty in assessing** learning outcomes due to **subjective judgments**
- Parents may be skeptical.

Competency based

'COMPETENCY' is the word used normally to describe a set of related skills.

- **THE AIM OF THE COMPETENCY BASED CURRICULUM IS TO:**
  - Make students self-motivated, robust and life-long learners
  - Strengthen the learning culture in the school
  - Improve results

Competency…

- Focus here is on competencies that will help in professional field. The different competencies for instance can be:
  - Cultural competencies
  - Community organizing
  - Data analysis and report writing
  - Conflict management
  - Project management
  - Proposal writing etc.

SPICES model of curriculum design

- Harden et al published an article in 1984 which was a seminal one in medical education.

Harden R, Sowden S and Dunn W (1984) Educational strategies in curriculum development: the SPICES model. *Medical Education*, 18:284-297.

- It described **six dimensions** around which a medical curriculum could be organised.

- Each of the dimensions could be regarded as a **continuum.**

- Thus a medical curriculum could be located somewhere along the dimensions of:
  - **S**tudent–centred
  - **P**roblem–based
  - **I**ntegrated
  - **C**ommunity–based
  - **E**lectives
  - **S**ystematic

Before SPICES

- Flexner (1910)
  - **T**eacher centred
  - **K**nowledge giving
  - **D**iscipline led
  - **H**ospital oriented
  - **S**tandard programme
  - **O**pportunistic (apprenticeship)

SPICES

- *Harden et al (1984)* **The SPICES model**
  - **S**tudent-centred
  - **P**roblem based
  - **I**ntegrated
  - **C**ommunity oriented
  - **E**lectives (+ core)
  - **S**ystematic

After SPICES

- Bligh et al (2001) **PRISMS**
  - **P**ractice based linked with professional development
  - **R**elevant to students and communities
  - **I**nter-professional and interdisciplinary
  - **S**horter courses taught in smaller units
  - **M**ultisite locations
  - **S**ymbiotic (organic whole)

Details of SPICES model

- **The purposes of the SPICES model are:**
  - to **review** an existing curriculum to see where improvements can be made
  - to develop a **new curriculum from scratch**
  - to **tackle specific questions** or issues relating to a curriculum
  - to help **decide** on what sort of **teaching methods** to use on a course
  - decide what **format the assessment** should take

SPICES…

- **S**tudent Centered
  - Most of the **emphasis** is on the **learner**
  - Increases learner **motivation**
  - Prepares learners for lifelong **self directed** education

SPICES…

- **P**roblem based
  - Helps develop **problem solving skills**
  - **Active participation** of the learner
  - Helps **deal with an overcrowded curriculum**; highlights **concepts** rather than knowledge

SPICES…

- Integrated
  - **Reduces fragmentation** of courses
  - Promotes staff **communication** and **collaboration** as departments are being integrated
  - **Holistic** approach to learning

SPICES…

- **C**ommunity oriented
  - Direct **contact** with the community the learners are being taught to serve
  - See a wide variety of **conditions** at a wide variety of **stages**
  - Learners learn about the **social and economic aspect of illness**

SPICES…

- **E**lectives
  - Allows curricular **flexibility**
  - Elective **+Core**
  - Enables learners to **explore further** interests in more details; matching to the learner's **needs**

SPICES…

- **S**ystematic
  - Teaching is structured and organized
  - Teaching is sequential
  - learner can become competent and confident with the least waste of time and resources
  - Easy for accreditation and validation

Bloom’s Taxonomy of learning domains

Introduction

- A committee of colleges, led by Benjamin Bloom (1956), identified three domains of educational activities:
  - **Cognitive**: Mental skills *(Knowledge)*
  - **Affective**: growth in feelings or emotional areas (*Attitude*)
  - **Psychomotor**: manual or physical skills (*Skills*)

Cognitive

- **It further includes six domains like**
  - knowledge,
  - comprehension,
  - application,
  - analysis,
  - synthesis and ,
  - evaluation

Knowledge

- Recalling specific and general items of information and also information about methods, processes and patterns

- **Key Words**: defines, describes, identifies, knows, labels, lists, matches, names, outlines, recalls, recognizes, reproduces, selects, states.

Comprehension

- Recognition of items of information settings similar to but different from those in which they were first encountered.

- **Key Words**: comprehends, converts, defends, distinguishes, estimates, explains, extends, generalizes, gives an example, infers, interprets, paraphrases, predicts, rewrites, summarizes, translates.

Application

- Explaining previously unseen data or events by applying knowledge from other situations

- **Key Words**: applies, changes, computes, constructs, demonstrates, discovers, manipulates, modifies, operates, predicts, prepares, produces, relates, shows, solves, uses.

Analysis

- Breaking down blocks of information into elements for the purpose of clarification.

- **Key Words**: analyzes, breaks down, compares, contrasts, diagrams, deconstructs, differentiates, discriminates, distinguishes, identifies, illustrates, infers, outlines, relates, selects, separates.

Synthesis

- Combining elements to form coherent units of information.

- **Key Words**: categorizes, combines, compiles, composes, creates, devises, designs, explains, generates, modifies, organizes, plans, rearranges, reconstructs, relates, reorganizes, revises, rewrites, summarizes, tells, writes.

Evaluation

- Making judgment about the value of information, materials or methods for given purposes

- **Key Words**: appraises, compares, concludes, contrasts, criticizes, critiques, defends, describes, discriminates, evaluates, explains, interprets, justifies, relates, summarizes, supports.

Affective

- The affective domain includes the manner in which we deal with things emotionally, such as
  - feelings,
  - values,
  - appreciation,
  - enthusiasms,
  - motivations, and
  - attitudes.

Affective…

- They are further divided into five levels:

Receiving: Sensitivity to certain stimuli and a willingness to receive or attend to them.

Key Words: asks, chooses, describes, follows, gives, holds, identifies, locates, names, points to, selects, sits, erects, replies, uses.

Affective…

- **Responding:** Involvement in a subject or activity or event to the extent of seeking it out, working with it or engaging in it.

- **Key Words**: answers, assists, aids, complies, conforms, discusses, greets, helps, labels, performs, practices, presents, reads, recites, reports, selects, tells, writes.

.

Affective…

- **Valuing:** Commitment to or conviction in certain goals, ideas or beliefs

- **Key Words**: completes, demonstrates, differentiates, explains, follows, forms, initiates, invites, joins, justifies, proposes, reads, reports, selects, shares, studies, works.

Affective…

- **Organisation:** Organisation of values into a system, awareness of relevance of and relations between appropriate values and the establishment of dominant personal values.

- **Key Words**: adheres, alters, arranges, combines, compares, completes, defends, explains, formulates, generalizes, identifies, integrates, modifies, orders, organizes, prepares, relates, synthesizes.

Affective…

- **Characterisation by a Value Complex:** Integration of beliefs, ideas and attitudes into a total philosophy of world view.

- **Key Words**: acts, discriminates, displays, influences, listens, modifies, performs, practices, proposes, qualifies, questions, revises, serves, solves, verifies.

Psychomotor domain

A person performing a demonstration

Introduction

- The psychomotor domain includes physical movement, coordination, and use of the motor-skill areas.
- Development of these skills requires practice
- measured in terms of speed, precision, distance, procedures, or techniques in execution.

Introduction…

- It talks about the third level of performance with the help of physical movement when the cognitive and affective levels have already met.

- It goes beyond the classroom setting and focuses on ability to apply the learnt things in practical life.

Introduction…

- The learner has reached the level to perform in front of other novice learners.
- Hence, the student has reached the level in which he is mentally prepared to act physically.
- Hence, the learner is able to use sensory cues (signals) to guide motor activity which gradually extends up to the level in which he can finally become used to it and further can develop new movement patterns.

Detail of the domain

- The different categories of the psychomotor domain from simple to complex level are given below:

1. Perception: The ability to use sensory cues to guide motor activity. Also called the selection step

  – **Key Words**: chooses, describes, detects, differentiates, distinguishes, identifies, isolates, relates, selects.

Categories…

2. Set: Readiness to act. It includes mental, physical, and emotional sets.

- **Key Words**: begins, displays, explains, moves, proceeds, reacts, shows, states, volunteers.

Categories…

3. Guided Response: The early stages in learning a complex skill that includes imitation and trial and error. Adequacy of performance is achieved by practicing.

- **Key Words**: copies, traces, follows, react, reproduce, responds

Categories…

4. Mechanism: This is the intermediate stage in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.

- **Key Words**: assembles, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches.

Categories…

5. Complex Overt Response:

- The skillful performance of motor acts that involve complex movement patterns.
- Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy.
- This category includes performing without hesitation, and automatic performance.
- The Key Words are the same as Mechanism, but will have adverbs or adjectives that indicate that the performance is quicker, better, more accurate, etc.

Categories…

6. Adaptation: Skills are well developed and the individual can modify movement patterns to fit special requirements.

Key Words: adapts, alters, changes, rearranges, reorganizes, revises, varies.

Categories…

7. Origination:

- Creating new movement patterns to fit a particular situation or specific problem.

- Learning outcomes emphasize creativity based upon highly developed skills.

Key Words: arranges, builds, combines, composes, constructs, creates, designs, initiate, makes, originates.